REMZI EFE DEMIR

Mechatronics Engineering Student | Embedded Systems, Power Electronics & UAVs

Electronics team lead with 2 years of hands-on autonomous vehicle development (UAV, AUV, USV); leading a GPS-denied swarm drone project approved by HAVELSAN and AFAD.

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HAVELSAN & AFAD	1st Prize	3 Conferences
Swarm drone project approved	Ankara University, wearable device	UMTİK'24 · MÜBAK'24 · MÜBAK'25

EDUCATION

Atılım University - Ankara, Türkiye · **B.Sc. Mechatronics Engineering** *Sep 2023 - Expected Jun 2027*

- Relevant coursework: Electronics, Microprocessors (PIC18 Assembly), Microcontroller Programming, Manufacturing Processes, Differential Equations

SKILLS

Electronics & Embedded: PCB design (Altium Designer), PCB assembly, power electronics / SMPS (flyback), LTspice, BMS design, STM32, ESP32, Arduino, Raspberry Pi, NVIDIA Jetson, Embedded C/C++, I2C / SPI / UART, RF communication

Autonomy & AI: Python, ROS, PID control, Kalman filter, reinforcement learning (Q-Learning, DQN), computer vision (OpenCV, YOLOv5), swarm intelligence, sim-to-real transfer, Linux

Mechanical & Process: SolidWorks, Siemens NX, 3D printing, MIL-STD-810, MIL-STD-498, SOP, Lastenheft (requirements specification), IFS ERP

Languages: Turkish (Native) • English (Professional) • Danish (Basic)

EXPERIENCE

Neutronix Technologies - Remote · **Voluntary Intern, Power Electronics** *Aug 2026 - Present*

- Designing a flyback switched-mode power supply: converter simulation in LTspice, schematic and PCB layout in Altium Designer.
- Assembling and testing PCB prototypes with STM32-based control; preparing SOP and Lastenheft (requirements specification) documents.

MZD Elektrik Elektronik A.Ş. · **Mechatronics Engineering Intern** *Jul 2026 - Aug 2026*

- Rotated through 4 departments of a defense cable-harness manufacturer and ROKETSAN strategic partner over 20 working days.
- Followed production planning end to end in IFS ERP, from purchasing work order to shipment, and studied shop-floor SOPs.

JupiterOcean Team · **Electronics Team Leader** *Sep 2024 - Aug 2026*

- AUV: designed power electronics, custom circuitry and battery system; led PCB design, sensor integration and communications; implemented Kalman-filtered IMU and PID orientation control; anomaly detection with OpenCV and YOLOv5. Presented at UMTİK'24 and MÜBAK'24.
- USV: designed electronic architecture and schematics in Altium Designer and the RF link between the USV and kamikaze drones; presented at MÜBAK'25.

QuadRobotics R&D and Software · **Voluntary Intern** *Jan 2025 - Mar 2025*

- Supported payload integration on the ROKETSAN KOZ quadruped platform with SolidWorks CAD; checked design compliance with MIL-STD-810 and MIL-STD-498.
- Contributed to payload hardware integration on robot dog platforms, NACA-based airfoil design and testing of autonomous robotic systems.

PROJECTS

Swarm Drone - GPS-denied indoor swarm · **Project Leader • 7-person team** *Jun 2025 - Present*

- Leading a 7-person team building a low-cost drone swarm for GPS-denied indoor environments; project approved by HAVELSAN and AFAD.
- Built the flight stack in-house: 3D-printed SolidWorks airframe, Altium flight-controller carrier PCB with power distribution, handheld RC with CRC-checked RF telemetry and own PID flight controller on ESP32.
- Training our own AI model for autonomous flight; currently in sim-to-real transfer, moving from a single-drone demo to the full swarm.

Autonomous Tracking & Moving-Platform Landing UAV · **Project Leader • 6-person team** *2026 - Present*

- Leading a 6-person team developing a UAV that autonomously tracks a target and takes off from and lands on a moving platform; completion targeted for 2027.

Wearable Health Tracking Device for Dysautonomia Patients · **Embedded Software Developer** *2024 - 2025*

- Implemented I2C, SPI and UART sensor data acquisition; the project won 1st prize at Ankara University.